

LEUKEMIAS/LYMPHOMAS

Leukemias are malignant diseases of myeloid or lymphoid cells which begin in the bone marrow.

Lymphomas are malignant diseases of lymphoid cells (NOT myeloid) that typically begin in lymph nodes

However, there is overlap as explained in the ScholarRx Bricks

- **All ACUTE leukemias have >20% blasts in bone marrow or peripheral blood**
- **Leukemias may be myeloid, lymphoid, chronic, or acute (CML, CLL, AML, ALL)**
- **It is important to understand hematopoiesis for myeloid v lymphoid – please see separate handout**
- **One way to view all these diseases is depicted on the next page, Fig. 1, taken and edited from Scholar Rx Bricks. THIS IS VERY OVERWHELMING so just use it as a guide to get through the material. You are not responsible for all these entities. Please see my comments throughout this handout.**
- **I have put red stars next to topics that are particularly important, and you will notice that under the 5 classifications of AML, you are only responsible for the first one.**
- **Also, all of Hodgkin's lymphoma is important, which is why there is a red circle around all the subtypes, but especially take note of the nodular sclerosis and (nodular) lymphocyte predominant, the latter of which is considered non-classical**
- **The blue ovals are important in the sense that you should not graduate medical school without knowing something about them, and don't ignore them just because they don't have red stars**

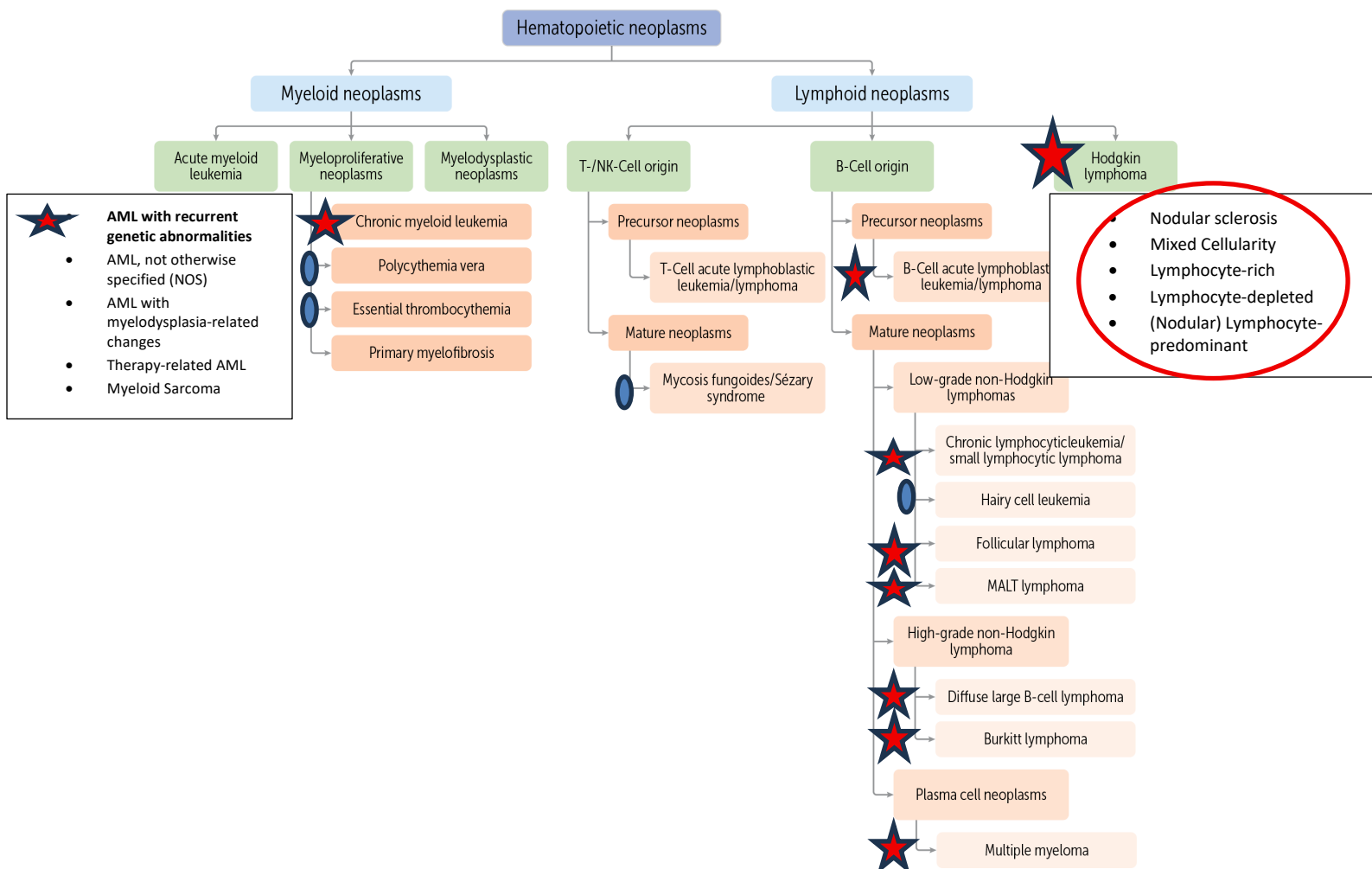


Figure 1 PLEASE NOTE UNDER ACUTE MYELOID LEUKEMIA THERE ARE 5 CATEGORIES BASED ON THE WHO CLASSIFICATION - YOU ARE NOT RESPONSIBLE FOR THE FAB CLASSIFICATION WHICH IS HISTORICAL. ALSO, UNDER HODGKIN LYMPHOMA there are 5 subcategories, the first four are considered Classical.

-Leukemias usually show blood abnormalities, and lymphomas usually present with lymphadenopathy.

-Many of these diseases, especially acute leukemias, multiple myeloma, etc. present similarly because the bone marrow fills with malignant immature cells which out crowd the normal rbc's, platelets, and wbc (myeloid) precursors resulting in anemia (pallor, fatigue), petechiae (low platelets), and infections (the normal wbc's are not present to fight infections)

- **Acute myeloid leukemia (AML)**, especially acute promyelocytic leukemia considered under the AML with recurrent genetic abnormalities classification – this is the only one you are responsible for!
 - ☐ **Acute promyelocytic leukemia (APL) translocation (15;17)**
 - ☐ **Auer Rods**
 - ☐ **May be associated with DIC (disseminated intravascular coagulation)**
 - ☐ **Tends to have a better prognosis; Rx All-Trans Retinoic Acid**
 - ☐ **Markers: Esterase + Tdt -**

- **Chronic myelogenous (myeloid) leukemia (CML)**
 - ☐ this is a **MYELOPROLIFERATIVE** Disorder (MPD) (along with Polycythemia Vera (PV), Essential Thrombocytosis (ET), and Myelofibrosis (MF)) - do NOT confuse with MYELODYSPLASIA, which you are not responsible for, except to know that myelodysplasias are usually pre-leukemic and usually present with cytopenia. The myeloproliferative disorders MAY lead to acute leukemia, but not as likely.
 - ☐ Philadelphia chromosome t(9;22) CML only - but sometimes in Adult ALL, which is rare
 - ☐ Will see hypercellular marrow showing **all levels of myeloid maturation**; Blasts 10-20% (normal blasts in bone marrow around 1 – 2%. NO blasts should be present in the peripheral blood)
 - ☐ **Note: all MPDs have increased basophilia** as well as increased neutrophils and myeloid precursors, PV usually has more increased rbcs, ET usually has more increased platelets; PV may be associated with pruritic; primary myelofibrosis means the bone marrow is replaced with fibrosis

- **Acute lymphoblastic leukemia (ALL)**
 - ☐ Mostly pediatric, though adults may rarely get ALL (with Philadelphia Chromosome). ALL in children is NOT usually associated with the Philadelphia chromosome t(9;22)
 - ☐ **Mostly B-cell** but T-cell ALL usually presents as a mediastinal mass in an adolescent male, not as common
 - ☐ Pallor (anemia), petechiae (thrombocytopenia), frequent infections, because even though the WBC is greatly elevated, there are mostly blasts and not mature cells
 - ☐ Sometimes bone pain (marrow expands)
 - ☐ May be associated with Down Syndrome
 - ☐ Lymphoblasts are Tdt + (versus myeloblasts which are negative as in AML above)

- **Chronic lymphocytic leukemia/lymphoma (CLL) – B cells**
 - ☐ Usually older population (>50-60), presenting with lymphocytosis, lymphadenopathy, hepatosplenomegaly
 - ☐ **Smudge cells** on peripheral smear
 - ☐ Not many specific translocations
 - ☐ **CD19, CD20, CD 23, and CD5**
 - **Multiple Myeloma (MM) : plasma cell neoplasm (there are others discussed in Bricks, you are not responsible for at this time)**
 - ☐ **monoclonal protein, M spike, seen best on serum electrophoresis (SPEP)**
 - ☐ >10% plasma cells, plasmablasts (bone marrow)
 - ☐ CRAB: hyperCalcemia, Renal dysfunctions, Anemia, Bone lesions)
 - Hairy Cell Leukemia – not responsible for but important for Boards; mostly males, usually extensive splenomegaly, “dry” bone marrow tap because the “hairs” on the surface of the cells get enmeshed, historically TRAP positive (immunohistochemical stain), very good prognosis
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- Note: Most lymphomas are B cell lymphomas: Hodgkins v. Non-Hodgkins lymphoma (NHL)
 - NHLs include Follicular cell lymphoma, diffuse large B cell lymphoma, Burkitt’s lymphoma, Marginal cell lymphoma, Mantle cell lymphoma (all of which come from B cells), and T cell lymphomas (much less common)
 - **Follicular Lymphoma** – B cell lymphoma; origin from germinal center cells within follicles of lymph nodes
 - CD markers 19 and 20, and others (but not under 10, those are T cell markers)
 - **Translocation t;(14;18)**
 - **Overexpression of Bcl-2**
 - Usually low grade but 10% may transform to diffuse large B cell lymphoma (which can also happen in CLL)

Diffuse Large B cell Lymphoma (DLBCL) – B cell, much more aggressive, advanced than follicular lymphoma, may be extranodal (outside lymph nodes)

- **Burkitt's lymphoma – B cell lymphoma**

- ☐ **High grade, extremely proliferative**
- ☐ **t(8;14) - leads to overexpression of MYC, resulting in excessive cell growth and division**
- ☐ Starry sky appearance of lymph nodes due to pale-staining macrophages (tingible body macrophages) eating dead/dying tumor cells (apoptosis), scattered among a dense, dark background of rapidly growing cancer cells.
- ☐ 3 forms – sporadic, endemic, immunosuppression related (esp. HIV associated)
- ☐ Sporadic form more common in North America and tends to be abdominal
- ☐ Endemic form is more common in Africa and tends to be mandibular (jaw mass), also associated with EBV
- ☐ Responsive to chemotherapy

- **Marginal cell lymphoma – B cell lymphoma**

- ☐ MALTomas (mucosa associated lymphoid tissue)
- ☐ **Helicobacter pylori infection in stomach** (causes gastritis, ulcers, and adenocarcinoma) may be associated with MALToma development; malignant small lymphocytes may invade glands (lymphoepithelial lesions)
- ☐ Rx with antibiotics
- ☐ Associated with other chronic diseases such as autoimmune diseases
- ☐ Usually good prognosis

- **Mantle cell lymphoma – B cell lymphoma, rare, just know: bcl-1 (NOT bcl-2 as in follicular), cyclin D1 overexpression, t(11;14)**

- **Mycosis Fungoides/Sezary Syndrome (skin involvement) – T cell lymphoma**

- ☐ CD4+ T cells with “cerebriform” nuclei
- ☐ Stages of skin involvement

- **Adult T cell leukemia/lymphoma (skin rash)- T cell lymphoma – just be aware of its existence, also a T cell lymphoma which affects the skin and blood mainly**

- **Hodgkin lymphoma – tends to spread in contiguous fashion v. non-contiguous for NHL. HL usually has a better prognosis than any NHL**
 - **Reed-Sternberg cells (come from B cells) – diagnostic cell**
 - **Variants of RS cells, EBV associations, CD 15 and CD 30 markers**
 - 5 subtypes of HL; first four **are positive for CD15 and CD30**, but negative for CD20 (and CD45)
 - Nodular sclerosis is the most common subtype; usually good/best prognosis, RS variant known as “lacunar” cell; collagen bands within LN
 - Mixed-cellularity – still good prognosis
 - Lymphocyte-rich – rare; good prognosis, depending on stage
 - Lymphocyte-depleted – **highest association with EBV**
 - Non-classical (nodular) lymphocyte-predominant HL (please note this is different than Lymphocyte-rich) has the **opposite phenotype as the 4 classical ones : CD15 negative and CD30 negative (but positive for CD20 and CD45)** ; usually excellent prognosis; the RS variant is known as the L and H or LP or “popcorn” cell

For all lymphomas, “B symptoms” refer to weight loss, night sweats, fevers, etc.

Review of Important translocations and markers:

- Tdt (lymphoblasts)
- Esterase (Myeloblasts)
- t (15;17) Acute promyelocytic leukemia
- t (9; 22) Philadelphia chromosome CML (minority of cases in Adu;t ALL)
- t (8;14) Burkitt lymphoma
- t(14;18) follicular lymphoma
- CD 15 and CD 30 Hodgkin disease; Classical v. Non-classical – see above
- CD 19, CD20, CD 23, CD 5 Chronic lymphocytic leukemia
- B cell CD markers v. T cell CD markers
- Bcl-2 follicular lymphoma
- Cyclin D-1 (bcl-1) mantle cell lymphoma t(11;14); bcl-1
- Most CD markers less than 10 are T cell markers

Also note that EBV is associated with the **endemic** form of Burkitt’s lymphoma and with some forms of Hodgkins lymphomas, especially lymphocyte-depleted

HEMORRHAGE/THROMBOSIS – this is not as detailed as leukemia and lymphoma because nomenclature is clearer and diseases are more straightforward. I've only highlighted major topics. The ScholarRx Bricks for these topics are optional. Please listen to the posted videos I recorded. Be familiar with these, especially:

- Acute v. Chronic Immune thrombocytopenic purpura
- Thrombotic thrombocytopenic purpura (pentad. ADAMST13 deficiency)
- Hemolytic Uremic Syndrome Typical v. Atypical
- Microcytic angioathic hemolytic anemia (schistocytes)
- Hemophilia A (X-linked inheritance, etc.)
- von Willebrand disease (do not need to know subtypes)
- Disseminated intravascular coagulation (presentation, labs, etc.)
- Factor V Leiden mutation (NOT deficiency) – primary (hereditary) causes of thrombosis
- Lines of Zahn, post mortem clots v. real thrombus
- Secondary causes of thrombosis (e.g., prolonged bed rest, obesity, pregnancy, etc.)
- Deep vein thrombosis and pulmonary embolism associations
- Bernard Soulier v. Glanzmann (platelet adhesion v. aggregation) – these are platelet qualitative disorders
- Important labs including PT/PTT, platelet count, bleeding time, d-dimers, etc.